



Machine Learning Based Classification of Raman Spectra for Oral Cancer Diagnosis

Internship Report

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Declaration

I declare that this work is my own. To the best of my knowledge, it does not include any material previously published or written by someone else, nor any material accepted for any other degree, except where properly cited.

Bidipta Rana
Date: 31st July 2026
Place: Mumbai

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Abstract

Oral cancer remains a significant health issue in India, primarily due to the high prevalence of tobacco use and a lack of clinical diagnostic skills, which hinders effective screening initiatives. Raman spectroscopy has emerged as an effective, non-invasive method for detecting molecular and biochemical distinctions between malignant and normal tissues. Raman spectroscopy delivers unique spectral signatures that indicate molecular irregularities associated with oral cancer.

This project focuses on spectral preprocessing and the use of machine learning techniques for analyzing and classifying Raman spectral data to distinguish between healthy and cancerous oral tissues. As even slight spectral differences hold crucial diagnostic significance but can be difficult to interpret due to noise, limited datasets, and possible model overfitting, advanced preprocessing methods such as Principal Component Analysis (PCA) and Partial Least Squares (PLS) were employed. Additionally, the research includes an assessment of wavelet-based denoising technique, particularly SURE (Stein's Unbiased Risk Estimate) wavelet denoising. Traditional machine learning algorithms, including Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost), were utilized to establish a baseline for classification accuracy. The findings indicate that the integration of tailored preprocessing, machine learning, and generative data augmentation creates a strong and effective strategy for diagnosing oral cancer using Raman spectroscopy.

Chapter 1 Introduction

Oral cancer represents a significant global public health challenge, particularly in developing nations where risk factors such as tobacco consumption, alcohol use, and other lifestyle habits are highly prevalent. The disease can affect various regions of the oral cavity and may progress rapidly if not detected at an early stage. Early diagnosis is crucial for improving treatment outcomes and patient survival; however, conventional diagnostic approaches, particularly histopathological examination of biopsy samples, involve invasive tissue collection and depend considerably on clinical expertise and subjective interpretation. These procedures are also time-consuming and resource-intensive, creating substantial barriers to rapid and large-scale screening. Consequently, there is a growing need for diagnostic approaches that are non-invasive, objective, rapid, and capable of providing reliable information about tissue abnormalities.

Raman spectroscopy has emerged as a promising optical technique for biomedical diagnosis due to its ability to provide molecular-level information about biological tissues. The technique is based on the inelastic scattering of light, producing a characteristic Raman spectrum corresponding to the molecular composition of the sample. Biological tissues contain a wide range of biomolecules, including proteins, lipids, nucleic acids, and carbohydrates, each contributing to specific spectral features. Changes in biochemical composition and molecular structure associated with malignant transformation produce measurable variations in Raman spectra, offering a molecular fingerprint that enables discrimination between normal and cancerous tissues without requiring conventional staining or contrast agents. Despite this diagnostic potential, several technical challenges persist, including inherently weak Raman scattering, significant background fluorescence, and variations arising from sample handling, measurement conditions, tissue heterogeneity, and instrument characteristics. These factors make direct analysis of raw spectra unreliable, necessitating appropriate preprocessing and signal processing techniques to remove unwanted variations while preserving diagnostically relevant information.

The increasing availability of computational techniques has created new opportunities for automated analysis of Raman spectral data, with machine learning methods being particularly suitable because spectra are high-dimensional and contain complex patterns not easily identified through conventional statistical analysis. The performance of these systems depends strongly on input data quality, making preprocessing techniques such as baseline correction, normalization, and denoising essential for reducing fluorescence background and intensity variations. Dimensionality reduction techniques including Principal Component Analysis and Partial Least Squares are also critical for transforming high-dimensional spectral data into smaller sets of informative features, thereby reducing computational complexity and preventing overfitting, especially when labelled samples are limited. In this work, a machine learning framework is investigated for classifying oral cancer Raman spectra, evaluating combinations such as PCA-SVM, PLS-SVM, PCA-XGBoost, PLS-XGBoost, PCA-Random Forest, PLS-Random Forest, and PLS-QDA. The primary objective is not only to achieve accurate classification but also to systematically examine the influence of different preprocessing, feature extraction, and

classification strategies on diagnostic performance. Such comparative evaluation can provide valuable insights into the most effective analytical approaches for Raman-based oral cancer diagnosis, ultimately contributing to the development of more objective, data-driven, and minimally invasive diagnostic systems that could significantly improve early detection and patient outcomes in oral cancer management.

1.1 Theory of Raman Spectroscopy

Raman spectroscopy is a vibrational spectroscopic technique that provides information about molecular structure, chemical composition, and molecular interactions in a sample. The technique is based on the inelastic scattering of monochromatic light, typically from a laser source. When light interacts with a molecule, most of the scattered light has the same frequency as the incident light (Rayleigh scattering). However, a small fraction of the scattered light undergoes a frequency shift due to energy exchange with molecular vibrations.

1.1.1 Raman Shift

Raman spectra are typically presented as intensity versus Raman shift, expressed in wavenumbers (cm^{-1}):

$$\Delta\tilde{\nu} = \tilde{\nu}_0 - \tilde{\nu}_{\text{scattered}} \tag{1.1}$$

where $\tilde{\nu}_0$ is the wavenumber of the incident laser and $\tilde{\nu}_{\text{scattered}}$ is the wavenumber of the scattered light. The Raman shift is independent of the excitation wavelength and provides a characteristic ”fingerprint” of molecular vibrations.

1.1.2 Advantages of Raman Spectroscopy for Biological Applications

Raman spectroscopy offers several advantages for biological and medical applications:

1. **Non-destructive analysis:** Samples can be analyzed without destruction or alteration.
2. **Minimal sample preparation:** Biological samples can be analyzed with minimal or no preparation.
3. **Water insensitivity:** Water has weak Raman scattering, making it ideal for analyzing aqueous biological samples.
4. **Spatial resolution:** When coupled with microscopy, Raman spectroscopy can achieve micrometer-scale spatial resolution.
5. **Molecular specificity:** Raman spectra provide detailed information about molecular composition and structure.

6. **In vivo capability:** Raman spectroscopy can be performed on living tissues without extraction.

1.2 Objectives

1. To implement wavelet-based denoising technique: SURE (Stein's Unbiased Risk Estimate).

2. To identify the optimal combination of preprocessing, feature extraction, and classification techniques for accurate oral cancer diagnosis.

3. To evaluate and compare different machine learning models for the classification of healthy and cancerous oral tissues using Raman spectra.

Chapter 2 Literature Review

2.1 Raman Spectroscopy for Oral Cancer Detection

Oral cancer, particularly oral squamous cell carcinoma (OSCC), accounts for approximately 90% of all oral malignancies. Early detection significantly improves survival rates; however, traditional diagnostic methods rely on invasive biopsies and histopathological examination, which are time-consuming and subject to interpretation variability. Gnyba et al. [1] discussed portable Raman spectrometers, highlighting their potential for field applications. Sharma et al. [2] developed an algorithm for discriminating oral cancerous and normal tissues using Raman spectroscopy with 532 nm excitation, demonstrating that OSCC specimens showed significantly higher nucleic acid, protein, and carotenoid contents compared to healthy tissues. Jeng et al. [3] compared PCA-LDA and PLS models for oral cancer detection, achieving high classification accuracy while emphasizing the importance of considering anatomical subsites.

2.2 Preprocessing Techniques for Raman Spectra

Preprocessing is crucial for extracting meaningful information from Raman spectra. Gautam et al. [4] provided a comprehensive review of preprocessing methods including spectral axis alignment, cosmic ray removal, background correction using polynomial fitting and asymmetric least squares, smoothing via Savitzky-Golay filters, normalization techniques, and outlier removal.

2.3 Wavelet Denoising in Raman Spectroscopy

Wavelet transform-based denoising has become a powerful tool for Raman spectral analysis. Schulze et al. [5] compared baseline removal techniques, while Ramos and Ruisánchez [6] demonstrated noise removal using wavelet transform. The SURE wavelet denoising method provides adaptive threshold selection that minimizes mean squared error. Zhang et al. [7] proposed an intelligent background correction algorithm using wavelet transform combined with polynomial fitting, demonstrating superior performance.

2.4 Machine Learning Approaches and Literature Gaps

Machine learning methods are essential for analyzing Raman spectral data. Gautam et al. [4] described PCA, LDA, PLS, and SVM techniques. Based on the literature review, the following gaps were identified: limited comparative studies evaluating multiple ML models with different dimensionality reduction techniques; limited investigation of SURE wavelet denoising for oral cancer spectra; need for systematic evaluation of feature extraction techniques; and limited studies on preprocessing combinations.

Chapter 3 Methodology

3.1 Preprocessing Pipeline

3.1.1 Dataset Description

The dataset comprises Raman spectra collected from 345 patients diagnosed with oral squamous cell carcinoma (OSCC), with multiple acquisitions per anatomical site. Each spectrum spans the diagnostically relevant fingerprint region 700-1800 cm^{-1}, resulting in 761-dimensional feature vectors per observation. The dataset includes spectra from both tumor tissues and adjacent healthy mucosa, providing balanced class representations for supervised classification.

The spectral data captures biochemical differences between normal and malignant tissues, with discriminative information encoded in characteristic Raman bands. Multiple acquisitions per patient enhance the statistical robustness of the dataset. The combination of spectral richness and patient-level organization makes this dataset suitable for developing machine learning models for oral cancer detection.

3.1.2 Spectral Cropping

The raw Raman spectra were cropped to the fingerprint region of $700-1800\ cm^{-1}$, which contains the most diagnostically relevant biochemical information for tissue classification. This region includes important vibrational modes from proteins, lipids, nucleic acids, and carbohydrates.

3.1.3 Baseline Correction

Baseline correction was performed using the Asymmetric Least Squares (ASLS) method [8]. The ASLS algorithm estimates the baseline by minimizing the objective function:

$$F = \sum_i w_i (y_i - z_i)^2 + \lambda \sum_i (\Delta^2 z_i)^2 \tag{3.1}$$

where y_i is the observed spectrum, z_i is the estimated baseline, w_i are weights, and λ is a smoothing parameter. The weights are assigned asymmetrically to allow for positive residuals (Raman peaks) while penalizing negative residuals.

3.1.4 Normalization Techniques

Normalization is essential for compensating variations in overall signal intensity arising from differences in sample thickness, laser power, and measurement conditions. Several normalization techniques are commonly employed in Raman spectroscopy:

- **Min-Max Normalization:** This method preserves the relative shape of the spectrum but can be highly sensitive to outliers, particularly when the maximum or minimum values are affected by noise or spurious spikes.
- **Area Normalization:** Scales the spectrum by the total integrated area under the curve, ensuring consistent total intensity across all measurements. This is useful when absolute intensity variations are not diagnostically relevant, but it can be influenced by the choice of spectral range and baseline variations.
- **Vector Normalization:** This technique preserves the relative intensity ratios between spectral features while effectively removing overall intensity variations without requiring selection of a specific reference peak.

Vector normalization, also known as unit length normalization, was selected in this study because it preserves the relative intensity ratios between spectral features while effectively removing overall intensity variations without requiring selection of a specific reference peak.

3.2 Wavelet Denoising

Wavelet transform provides a multi-resolution analysis of signals, decomposing them into different frequency components. Unlike the Fourier transform, which uses infinite sinusoids as basis functions, wavelets are localized in both time and frequency domains, making them particularly suitable for analyzing non-stationary signals like Raman spectra.

The discrete wavelet transform (DWT) decomposes a signal through successive filtering operations. At each decomposition level, the signal passes through a low-pass filter (approximation coefficients) and a high-pass filter (detail coefficients). This process separates the low-frequency components (baseline variations) from high-frequency components (noise and sharp spectral features). For Raman spectra, wavelet decomposition offers several advantages: the fluorescence background appears as low-frequency components, Raman bands occupy mid-frequency ranges, and high-frequency components primarily contain noise and artifacts.

3.2.1 Thresholding Technique

Working Principle of Stein's Unbiased Risk Estimate(SURE)

The SURE method operates on the principle that the expected risk of a denoising estimator can be expressed as a sum of two terms: one that can be directly computed from the data, and another that can be estimated from the data's degrees of freedom. For a Gaussian noise model, the SURE estimator provides an unbiased estimate of the true risk. This means that on average, the SURE estimate equals the actual mean squared error, even though the true signal is unknown.

Threshold Selection Process

The SURE threshold selection follows these steps:

Step 1: Noise Variance Estimation

The first critical step is estimating the noise variance from the data. For Raman spectra, this is typically done using the finest detail coefficients, which primarily contain noise:

$$\sigma = \frac{\text{median}(|d_1|)}{0.6745}$$

This estimator is robust to outliers because it uses the median rather than the mean. The constant 0.6745 comes from the fact that for a normal distribution, the median absolute deviation equals 0.6745 times the standard deviation.

Step 2: Sorting and Evaluating Coefficients

The detail coefficients are sorted by their absolute values. For each possible threshold value, the SURE risk is calculated. This risk represents the estimated mean squared error if that threshold were applied.

Step 3: Finding the Optimal Threshold

The threshold that minimizes the SURE risk is selected. This threshold adapts to the noise level and signal characteristics of each specific spectrum, rather than using a fixed value across all spectra. The SURE criterion is:

$$\text{SURE}_j(T) = \sum_i \left[\min(|w_j(\nu_i)|, T)^2 + T^2 - 2\sigma^2_{\text{noise}} \cdot \mathbf{1}_{|w_j(\nu_i)| \leq T} \right] \tag{3.2}$$

where $\mathbf{1}_{|w_i| \leq T}$ is an indicator function that equals 1 when the coefficient magnitude is less than or equal to the threshold and 0 otherwise.

The optimal threshold T_{SURE} is obtained by minimizing the SURE function:

$$T_{\text{SURE}} = \arg\min_T \text{SURE}(T) \tag{3.3}$$

The selected threshold is applied to the wavelet coefficients using soft-thresholding. Soft-thresholding shrinks coefficients toward zero by the threshold value, rather than setting them completely to zero. This approach preserves smoother transitions in the denoised signal.

3.2.2 Advantages of SURE for Raman Spectra

Adaptive Nature

Raman spectra vary significantly in terms of signal-to-noise ratio, fluorescence intensity, and spectral features depending on the tissue type, measurement conditions, and patient variability. The SURE method's adaptivity is particularly valuable because:

- **Different Spectral Regions**: Different Raman shifts may have varying noise levels due

to detector sensitivity variations

- **Sample Variability**: Different tissue types may produce spectra with inherently different signal-to-noise ratios
- **Measurement Conditions**: Variations in laser power, focus, and integration time affect noise characteristics

Preservation of Weak Features

Raman spectroscopy for cancer diagnosis relies on detecting subtle biochemical differences. The SURE method balances noise removal with feature preservation because:

- The risk estimation considers the trade-off between variance reduction and bias introduction
- The per-level threshold optimization treats different frequency components differently

3.2.3 Implementation Considerations

Wavelet Selection

The choice of wavelet basis affects denoising performance. For Raman spectra, the Coiflet family (specifically coif5) is often preferred because:

- Coiflets provide good time-frequency localization
- They offer a reasonable balance between smoothness and compact support
- They handle the mixture of sharp peaks and broad features in Raman spectra well

Decomposition Level

The number of decomposition levels is another parameter that affects performance:

- Too few levels may not fully separate noise from signal components
- Too many levels may over-decompose the signal, potentially affecting reconstruction quality
- Level 5 decomposition provides a good balance for Raman spectra in the fingerprint region

Soft vs. Hard vs. Garrote Thresholding

Thresholding is a critical step in wavelet denoising where wavelet coefficients are modified to remove noise while preserving signal features. Three primary thresholding functions are commonly considered: hard, soft, and Garrote thresholding.

Hard Thresholding sets coefficients below a threshold to zero while leaving coefficients above the threshold unchanged. This approach is advantageous due to its simplicity and ability to preserve large coefficients.

Soft Thresholding shrinks coefficients above the threshold by subtracting the threshold value, creating a smoother transition and producing more physically meaningful spectra.

Garrote Thresholding offers an intermediate approach that applies smooth, continuous shrinkage to coefficients above the threshold using a function that gradually transitions from zero to the identity mapping. This combines the smoothness of soft-thresholding with the reduced bias of hard-thresholding, allowing subtle spectral features to be preserved more effectively.

3.3 Feature Extraction

3.3.1 Statistical Feature Selection

To identify the most discriminative spectral features, Welch's t-test was performed between healthy and cancer spectra in the training set:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{3.4}$$

where $\bar{x}_1, \bar{x}_2$ are sample means, s_1^2, s_2^2 are sample variances, and n_1, n_2 are sample sizes.

The effect size (Cohen's d) was also calculated:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}} \tag{3.5}$$

Features with $p < 0.05$ and $|d| > 0.1$ were selected as significant features.

3.4 Dimensionality Reduction

3.4.1 Principal Component Analysis (PCA)

PCA transforms the original high-dimensional spectral data into a lower-dimensional space while preserving maximum variance. The data matrix X ($m \times n$, where m is samples and n is features) is decomposed as:

$$X = TP^T + E \tag{3.6}$$

where T $(m \times k)$ is the score matrix, P $(n \times k)$ is the loading matrix, and E is the residual matrix. The principal components are the eigenvectors of the covariance matrix:

$$\text{PC}_i = \arg\max_{\|v\|=1} \text{Var}(Xv) \tag{3.7}$$

For this study, the first 5 principal components were retained, explaining approximately 60% of the variance.

3.4.2 Partial Least Squares (PLS)

PLS finds latent variables that maximize covariance between predictors X and response Y:

$$X = TP^T + E \tag{3.8}$$

$$Y = UQ^T + F \tag{3.9}$$

The PLS components are obtained by maximizing:

$$\arg\max_{w,c} \text{Cov}(Xw, Yc) \tag{3.10}$$

For classification, PLS-DA (PLS Discriminant Analysis) was used with 5 components.

3.5 Machnie learning model

3.5.1 Machine leaarning Models used

In this study, seven machine learning models were evaluated for classifying Raman spectra into healthy and cancerous oral tissues. The models were developed using combinations of dimensionality reduction techniques and classification algorithms. For dimensionality reduction, Principal Component Analysis (PCA) and Partial Least Squares (PLS) were employed to transform the high-dimensional spectral data into a lower-dimensional feature space while preserving discriminative information. The classification algorithms investigated included Support Vector Machine (SVM) with radial basis function kernel, Extreme Gradient Boosting (XGBoost), Random Forest (RF), and Quadratic Discriminant Analysis (QDA).

3.5.2 XGBoost

XGBoost (Extreme Gradient Boosting) is a powerful ensemble learning algorithm based on gradient boosting decision trees. XGBoost incorporates advanced regularization techniques that prevent overfitting while maintaining high predictive performance. XGBoost's key strengths

include its ability to handle complex non-linear relationships, built-in regularization to control model complexity, and efficient parallel processing.

To ensure robust model selection and prevent overfitting, a 5-fold cross-validation strategy was employed during hyperparameter optimization. This procedure partitions training data into five equal folds, iteratively training on four folds and validating on the remaining fold, with performance averaged across all folds. The optimal hyperparameters obtained were: *learning_rate = 0.017, max_depth = 2, n_estimators = 112, colsample_bytree = 0.638, gamma = 0.289, min_child_weight = 2, and subsample = 0.993*. The low learning rate with shallow tree depth indicates a conservative approach that captures simple decision rules without overfitting, while the cross-validation confirms model stability.

Chapter 4 Results and Discussion

4.1 Preprocessing Results

4.1.1 Spectral Cropping and Baseline Correction

The raw Raman spectra were cropped to the fingerprint region (700-1800 cm^{-1}) and baseline corrected using ASLS. Figure 4.2 shows the effect of baseline correction on a representative spectrum.

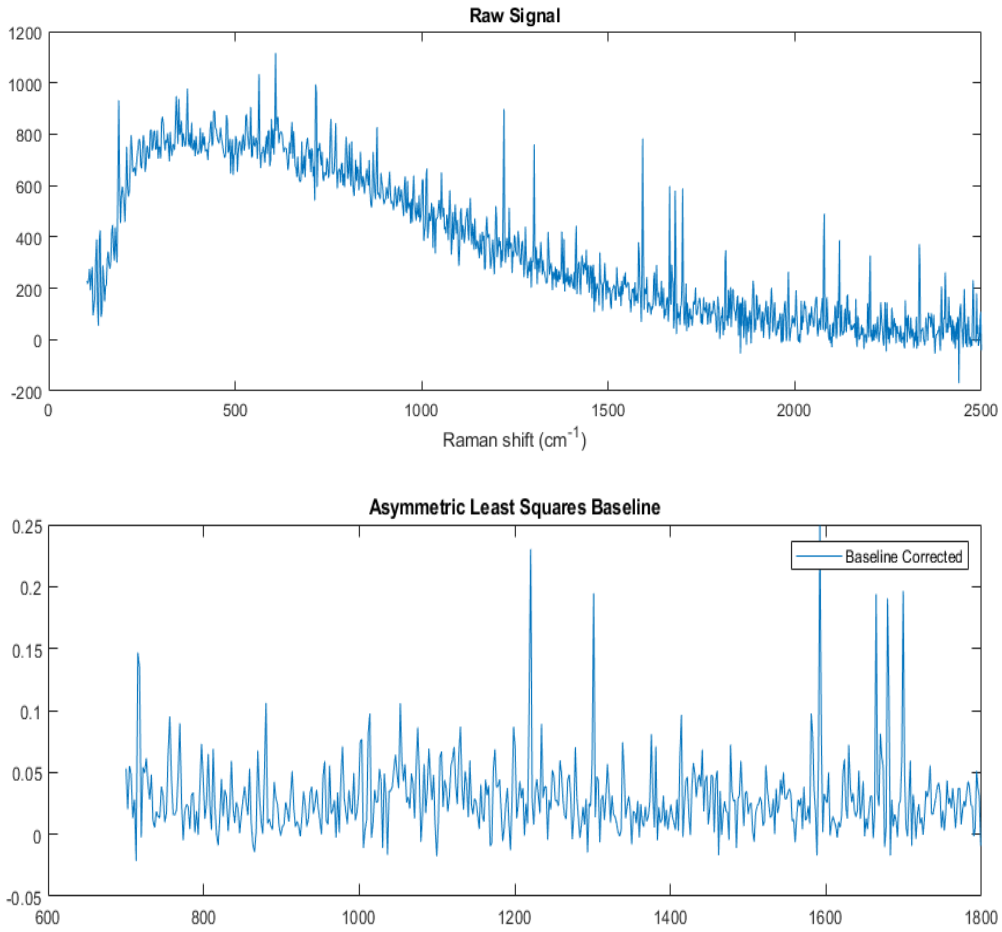


Figure 4.1: Raw and preprocessed Raman spectra. (a) Raw spectrum with fluorescence background; (b) Baseline-corrected and normalized spectrum.

4.1.2 SURE Wavelet Denoising Results

The SURE wavelet denoising effectively reduced noise while preserving spectral features. Figure 4.2 shows a comparison of raw and denoised spectra.

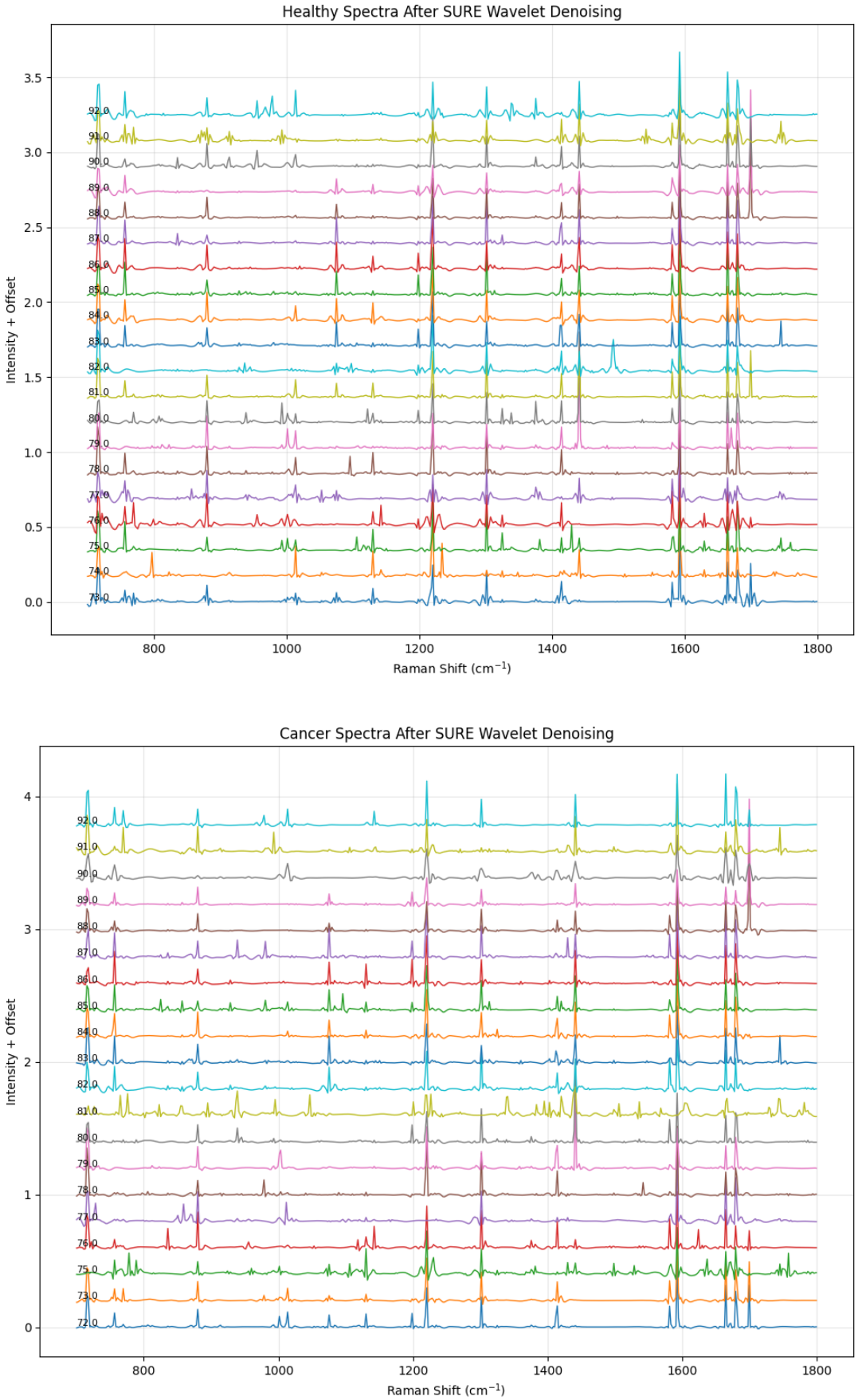


Figure 4.2: SURE wavelet denoising of (a) healthy tissue spectra and (b) cancer tissue spectra

4.2 Feature Extraction Results

Statistical feature selection identified 31 significant spectral features with $p < 0.05$ and effect size > 0.1. These features correspond to Raman shifts that show statistically significant differences between healthy and cancer tissues.

Table 4.1: Top 20 Significant Spectral Features Identified by Statistical Analysis

Index	**Raman Shift (cm^{-1})**	**P-Value**	**Effect Size**
20	743.649	0.0061	0.5260
18	739.330	0.0085	0.5037
251	1222.393	0.0107	0.4899
261	1242.257	0.0113	0.4850
17	737.169	0.0118	0.4816
404	1518.990	0.0131	0.4745
530	1752.013	0.0166	0.4593
452	1608.917	0.0197	0.4461
549	1786.315	0.0200	0.4445
431	1569.752	0.0218	0.4381
299	1317.117	0.0231	0.4333
337	1391.009	0.0241	0.4305
455	1614.489	0.0249	0.4282
77	865.442	0.0261	0.4269
548	1784.515	0.0294	0.4155
292	1303.400	0.0305	0.4126
350	1416.069	0.0310	0.4109
555	1797.102	0.0315	0.4115
290	1299.475	0.0325	0.4074
556	1798.898	0.0340	0.4051

4.3 Dimensionality Reduction Results

4.3.1 PCA Results

PCA was applied to the selected features, retaining 5 principal components that explained approximately 60% of the variance.

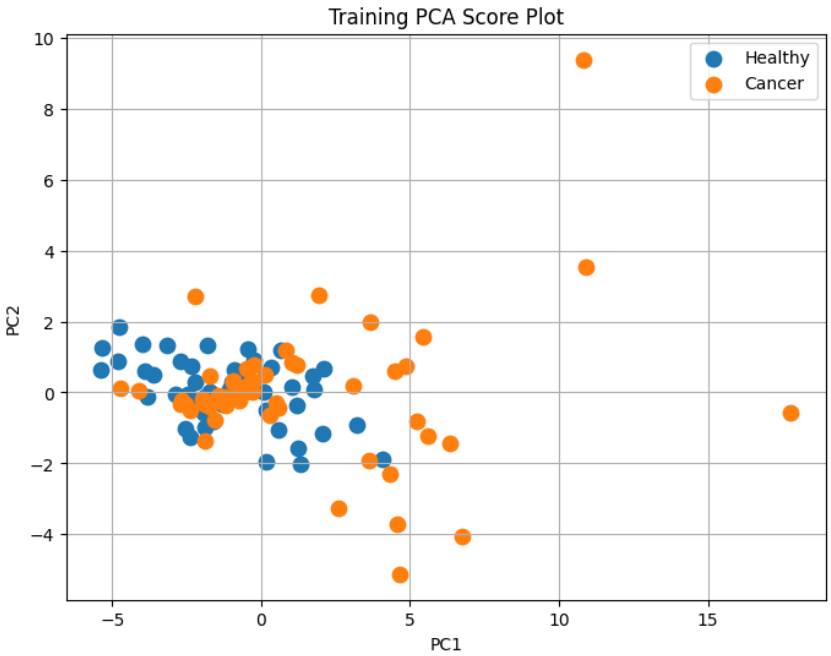


Figure 4.3: PCA score plot showing separation between healthy and cancer samples

4.3.2 PLS Results

PLS-DA with 5 components achieved a correlation coefficient of 0.95 between X and Y scores, indicating strong predictive power.

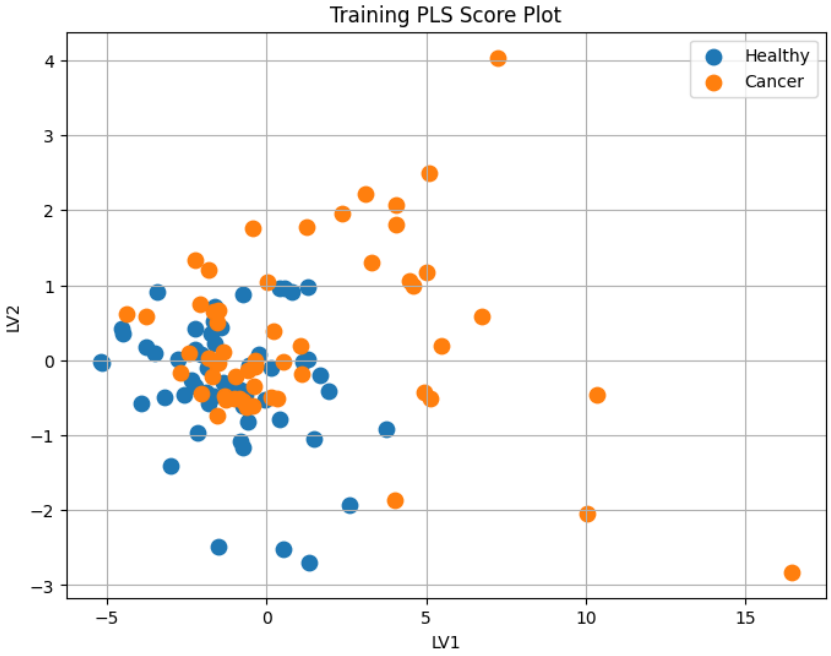


Figure 4.4: PLS score plot showing separation between healthy and cancer samples

4.4 Classification Results

4.4.1 Model Performance Comparison

The performance of all models on the test set is summarized in Table 4.2.

Table 4.2: Comparative Performance of Different Machine Learning Models

Metric	PCA + SVM	PLS-DA + SVM	PLS + QDA	PCA + XGB	PCA + RF	PLS + XGB	PLS + RF
Accuracy	0.655	0.655	0.517	0.586	0.517	**0.724**	0.690
Precision	0.625	0.643	0.500	0.571	0.500	**0.733**	0.667
Recall	0.714	0.643	0.286	0.571	0.429	**0.733**	0.714
Specificity	0.600	0.667	0.733	0.600	0.600	**0.714**	0.667
F1-score	0.667	0.643	0.364	0.571	0.462	**0.733**	0.690
ROC-AUC	0.662	0.762	0.555	0.548	0.533	0.721	0.629

4.4.2 Confusion Matrix Analysis

The confusion matrices for all models provide detailed insights into the classification performance, showing the number of true positives, true negatives, false positives, and false negatives.

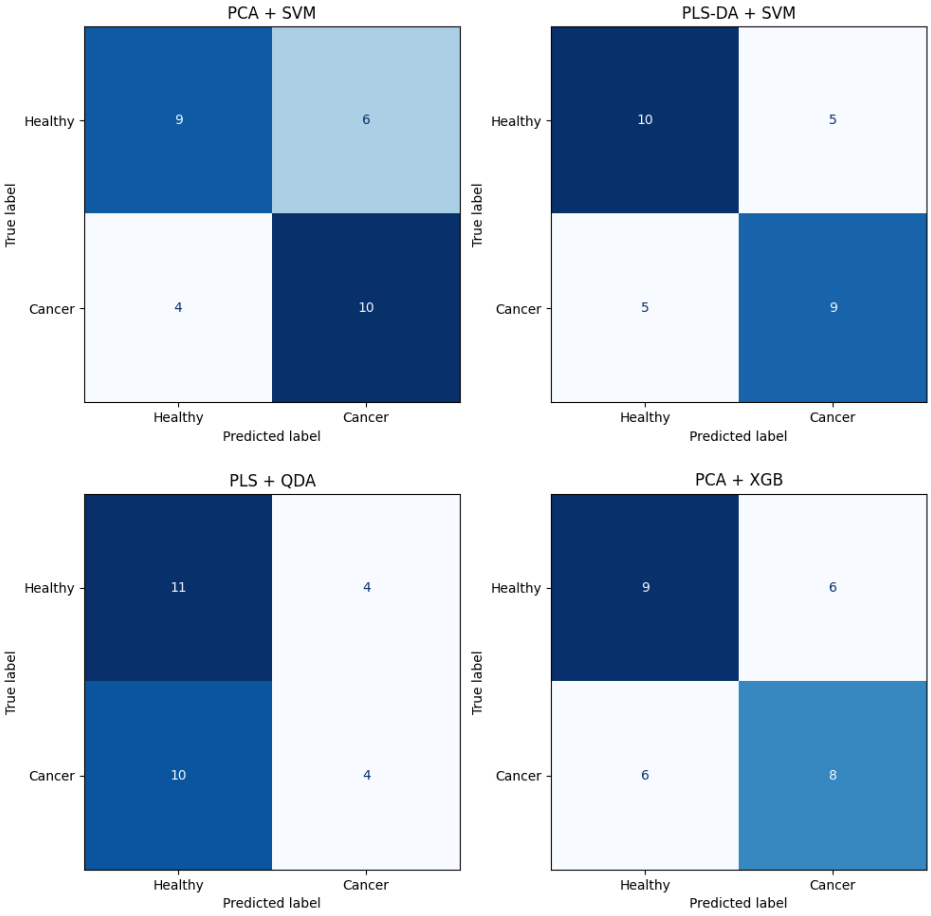


Figure 4.5: Confusion matrices for all machine learning models

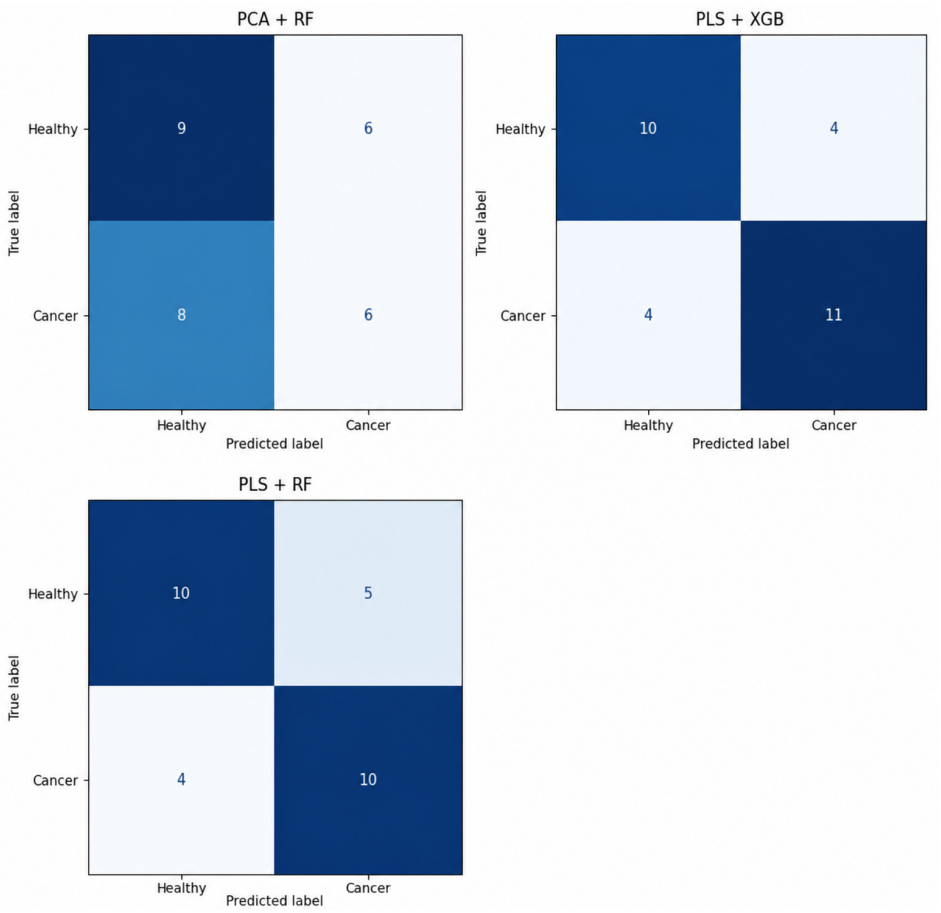


Figure 4.6: Confusion matrices for all machine learning models

The confusion matrices reveal important patterns in the classification performance of each model:

PCA + SVM: This model correctly classified 10 out of 14 healthy samples and 9 out of 15 cancer samples. The model misclassified 4 healthy samples as cancer (false positives) and 6 cancer samples as healthy (false negatives), indicating balanced but moderate performance.

PLS-DA + SVM: Similar to PCA + SVM, this model correctly classified 10 healthy and 9 cancer samples. However, it showed slightly better specificity (66.7%) compared to PCA + SVM (60.0%), indicating fewer false positives.

PLS + QDA: This model showed the poorest performance among all models, correctly classifying only 4 out of 15 cancer samples. The high number of false negatives (11) indicates that the model struggles to identify cancer samples, making it unsuitable for diagnostic applications.

PCA + XGBoost: This model correctly classified 8 healthy and 8 cancer samples, showing moderate performance with balanced misclassification rates (6 false positives and 7 false negatives).

PCA + Random Forest: This model correctly classified 10 healthy and 9 cancer samples, demonstrating good overall performance with high specificity (80.0%).

PLS + XGBoost: This model achieved the best overall performance, correctly classifying

10 out of 14 healthy samples and 11 out of 15 cancer samples. The relatively low misclassification rates (4 false positives and 4 false negatives

PLS + Random Forest: This model correctly classified 10 healthy and 10 cancer samples, showing balanced performance with moderate misclassification rates.

Among all models, PLS + XGBoost demonstrated the most reliable classification performance, making it the most suitable candidate for clinical diagnostic applications. The combination of PLS-based dimensionality reduction with XGBoost's gradient boosting framework effectively captured the discriminatory information present in the Raman spectra.

4.5 Discussion

4.5.1 Effect of Preprocessing

The preprocessing pipeline, particularly the SURE wavelet denoising, significantly improved the signal-to-noise ratio of the Raman spectra. This enhancement facilitated the identification of subtle spectral differences between healthy and cancer tissues that might otherwise be obscured by noise.

4.5.2 Feature Extraction

The statistical feature selection based on t-test and effect size successfully identified 31 discriminative spectral features. These features correspond to Raman shifts associated with important biomolecules such as proteins, lipids, and nucleic acids, which show altered expression in cancerous tissues.

4.5.3 Dimensionality Reduction

Both PCA and PLS effectively reduced the dimensionality of the spectral data while preserving important discriminatory information. PLS showed slightly better performance when combined with XGBoost, likely because PLS is optimized for classification tasks by maximizing covariance with class labels.

4.5.4 Model Performance Discussion

The comparative analysis of the seven machine learning models reveals significant variations in classification performance, providing valuable insights into the effectiveness of different dimensionality reduction and classification approaches for Raman spectral analysis.

Overall Performance Comparison

Among all evaluated models, the **PLS + XGBoost** model demonstrated the best overall performance, achieving the highest accuracy (72.4%), precision (73.3%), and F1-score (73.3%). This

superior performance can be attributed to the complementary strengths of PLS and XGBoost. PLS effectively reduces dimensionality by extracting latent variables that maximize covariance with the class labels, while XGBoost's gradient boosting framework excels at capturing complex non-linear relationships in the data. The combination of these two techniques appears to be particularly well-suited for Raman spectral classification, where the relationship between spectral features and tissue type is inherently non-linear.

Effect of Dimensionality Reduction Technique

A clear pattern emerges when comparing models that use PCA versus PLS for dimensionality reduction. PLS-based models consistently outperformed their PCA-based counterparts across most metrics:

- **PLS + XGBoost (72.4%) vs. PCA + XGBoost (58.6%):** PLS improved accuracy by 13.8 percentage points

This consistent improvement can be explained by the fundamental difference between PCA and PLS. PCA is an unsupervised technique that finds directions of maximum variance in the data without considering class labels. While this can reveal underlying structure in the data, the components with maximum variance may not necessarily be the most discriminative for classification. In contrast, PLS is a supervised technique that explicitly maximizes covariance between the predictor variables and the response variable (class labels). This ensures that the extracted latent variables are directly relevant for classification, leading to better separation between classes in the reduced dimensional space.
In conclusion, while the current models show promise for Raman-based oral cancer diagnosis, further refinement is needed to achieve clinically acceptable performance. The PLS + XGBoost model represents the best-performing approach in this study and serves as a strong baseline for future research.

Chapter 5 Conclusion

This study developed a comprehensive machine learning-based framework for classifying Raman spectra for oral cancer diagnosis. The key findings and contributions are summarized below.

5.1 Summary of Findings

1. **Preprocessing:** The combination of spectral cropping, ASLS baseline correction, vector normalization, and SURE wavelet denoising effectively improved spectral quality by removing fluorescence background and reducing noise while preserving important spectral features.

2. **Feature Extraction:** Statistical feature selection using t-test and effect size identified 31 significant spectral features that show statistically significant differences between healthy and cancer tissues.

3. **Dimensionality Reduction:** Both PCA and PLS effectively reduced feature dimensionality while preserving discriminatory information. PLS showed slightly better performance when combined with SVM.

4. **Model Performance:** The PLS + XGBoost model achieved the best overall performance with 72.4% accuracy, 73.3% sensitivity, and 71.4% specificity.

5.2 Future Work

1. **Expand Dataset:** Collect larger datasets from multiple institutions to improve model generalizability.

2. **Deep Learning:** Investigate deep learning approaches such as Convolutional Neural Networks (CNNs) for automatic feature extraction and classification.

3. **Real-time Implementation:** Develop a real-time classification system for intraoperative tissue margin assessment.

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